GUTCP Cosmological Model 2023: Audit of a Redshift Counterpart

June 9, 2026

Scope

This note audits the reconstructed 2023 Chapter 32 material in `docs/reconstructed.tex.pages/` against the existing Mathematica notebook `GUTCPCosmologicalModel.nb`. The target is not a FLRW redshift parameter. The target is the smallest GUTCP-native candidate for a wavelength shift relation that can be written from Mills's equations without importing $1+z=a(t_o)/a(t_e)$.

The notation below treats cosmological observables as relations. A relation may later become a singleton graph after branch, domain, and calibration constraints are added, but that uniqueness is extra information and should not be hidden inside a symbol. Thus expressions such as $H(t)$ or $d_A(z)$ are read as graph/predicate notation unless the text explicitly states that a singleton branch has been selected.

Equation Audit

Mills equation	Page	Section	Extracted TeX	Notebook symbol	Status
32.126	1536	Hubble law	$v = HR$	redshift path premise	supporting
32.36	1519	Gravity / particle production	$r_g = \frac{2Gm_0}{c^2}$	`2 G m/c^2`	matched
32.38	1520	Gravity metric	$\mathrm{d}\tau^2 = -\left(1-\frac{2Gm_0}{c^2r}\right)\mathrm{d}t^2 - \left(1-\frac{2Gm_0}{c^2r}\right)^{-1}\frac{\mathrm{d}r^2}{c^2} + \frac{r^2}{c^2}\mathrm{d}\theta^2 + \frac{r^2\sin^2\theta}{c^2}\mathrm{d}\phi^2$	implicit	supporting
32.43	1523	Particle production	$\tau = \frac{GM}{r^*c^2} = \frac{r_g}{c} = \frac{v_g}{c}t_i = t_i\Delta$	time metric	supporting, OCR risk
32.140	1542	Expanding universe	$Q = \frac{c^3}{4\pi G}$	`c^3/(4 Pi G)`	matched
32.146	1544	Minimum radius	$\Re_{\min} = \left(\frac{\cdots}{4\pi\sigma e T^4}\right)^{1/2}$ with printed result $8.52\times10^{27}\,\mathrm{m}$	`alephMin`	ambiguous norm.
32.149	1545	Oscillation period	$T_U = \frac{4\pi G m_U}{c^3}$, but the printed numerical value is $3.10\times10^{19}\,\mathrm{s}$ for $m_U \simeq 2\times10^{54}\,\mathrm{kg}$	period in `aleph[t_]`	mismatch: numeric convention uses $2\pi G m_U/c^3$

Mills equation	Page	Section	Extracted TeX	Notebook symbol	Status
32.153	1546	Radius evolution	$X(t) = 2.16 \times 10^{28} - 1.86 \times 10^{28} \cos\left(\frac{2\pi t}{9.83 \times 10^{11}\,\mathrm{yr}}\right)$ m	`aleph[t_]`	matched numeric
32.154	1546	Expansion rate	$\dot{X} = 4\pi c \sin\left(\frac{2\pi t}{9.83 \times 10^{11}\,\mathrm{yr}}\right)$	`alephRate[t_]`	matched
32.156	1547	Hubble constant	$H(t) = \frac{\dot{S}}{ct} = \frac{4\pi c \sin(\cdots)}{ct}$	`H[t_]`	matched; denominator under audit
32.158	1548	Matter inventory	$m_U(t) = \frac{m_U}{2}\left(1 + \cos\left(\frac{2\pi t}{2\pi G m_U/c^3}\right)\right)$	`mUBook[t_]`	matched; commentary conflict
32.165	1551	Temp./redshift	$\lambda_\infty = \lambda(r)\left(1 + \frac{2Gm_U}{rc^2}\right)$	`zGUTCP`	endpoint correction
p.1579	1579	CMBR structure scale	$L_\ell = \frac{2}{\ell}\, 2\pi r_{\mathrm{sphere}} = \frac{4\pi r_{\mathrm{sphere}}}{\ell}$	`cmbrScale`	stated; not BAO

GUTCP z: Denominator Audit

[FORCED] The observational redshift is defined by the wavelength ratio

$$1 + z = \frac{\lambda_o}{\lambda_e}.$$

Mills gives a Hubble-law structure in Eq. (32.126), $v = HR$, and then gives the oscillatory radius and rate in Eqs. (32.153)–(32.154). Page 1540–1544 frames the mechanism as mass-energy conversion causing spacetime expansion: matter creation contracts spacetime, matter-to-energy release expands it, and redshift/blueshift time stamps the extent of spacetime expansion between observer and observed. Therefore the denominator in $H = \dot{R}/R$ must be the spacetime scale or extent being compared.

Using the notebook and the numerical convention of Eqs. (32.153)–(32.156),

$$\theta(t) = \omega t, \qquad \omega = \frac{2\pi}{T_U}, \qquad T_U = \frac{2\pi G M_0}{c^3}, \tag{1}$$

$$\mathcal{R}(t) = \frac{2GM_0}{c^2} + \frac{4\pi G M_0}{c^2}\left[1 - \cos\theta(t)\right], \qquad \dot{\mathcal{R}}(t) = 4\pi c \sin\theta(t). \tag{2}$$

The printed Eq. (32.149) instead gives $T_U = 4\pi G M_0/c^3$; that factor-of-two choice remains a convention switch.

[CHOSEN] Mills's Eq. (32.156) replaces the Hubble-law denominator by ct:

$$H_{ct}(t) = \frac{\dot{\mathcal{R}}(t)}{ct} = \frac{4\pi \sin\theta(t)}{t}. \tag{3}$$

This is the step that creates a damped sinc-like graph from an oscillatory radius. Its associated path-redshift relation is

$$\boxed{\mathfrak{Z}^{ct}_{\mathrm{GUTCP}}(t_e, t_o, z) \Longleftrightarrow 1 + z = \exp\left[\int_{t_e}^{t_o} H_{ct}(t)\,\mathrm{d}t\right]} \tag{4}$$

or, for the numerical period convention,

$$\boxed{\mathfrak{z}^{ct}_{\mathrm{GUTCP}}(t_e,t_o,z) \iff 1+z = \exp[4\pi\,(\mathrm{Si}(\omega t_o)-\mathrm{Si}(\omega t_e))]\,,} \tag{5}$$

where $\mathrm{Si}(x)=\int_0^x(\sin u/u)\,\mathrm{d}u$. This preserves the notebook calculation, but it is not forced by the arrow-of-time construction.

[FORCED] If Eq. (32.126) is applied to the whole oscillatory GUTCP radius, then

$$H_{\mathcal{R}}(t)=\frac{\dot{\mathcal{R}}(t)}{\mathcal{R}(t)}, \qquad \mathrm{d}\ln(1+z_{\mathrm{GUTCP}})=H_{\mathcal{R}}(t)\,\mathrm{d}t.$$

The integral is exact:

$$\boxed{\mathfrak{z}^{\mathcal{R}}_{\mathrm{GUTCP}}(t_e,t_o,z) \iff 1+z=\frac{\mathcal{R}(t_o)}{\mathcal{R}(t_e)}.} \tag{6}$$

This expression has the same algebraic shape as a FLRW scale-factor ratio, but here it is derived from Mills's $v=HR$ and his $\mathcal{R}(t)$, not assumed from FLRW geometry.

[CHOSEN] The stronger reading of page 1542 is that redshift time stamps the *extent of spacetime expansion* from the minimum-radius bounce, not the entire radius including the finite gravitational-radius offset. Define

$$\mathcal{E}(t)=\mathcal{R}(t)-\mathcal{R}_{\min}, \qquad \mathcal{R}_{\min}=\frac{2GM_0}{c^2}. \tag{7}$$

Then

$$H_{\mathcal{E}}(t)=\frac{\dot{\mathcal{E}}(t)}{\mathcal{E}(t)}=\frac{\dot{\mathcal{R}}(t)}{\mathcal{R}(t)-\mathcal{R}_{\min}},$$

and the corrected expansion-extent redshift is

$$\boxed{\mathfrak{z}^{\mathcal{E}}_{\mathrm{GUTCP}}(t_e,t_o,z) \iff 1+z=\frac{\mathcal{R}(t_o)-\mathcal{R}_{\min}}{\mathcal{R}(t_e)-\mathcal{R}_{\min}}.} \tag{8}$$

This removes the artificial $1/t$ damping introduced by ct, and it keeps Mills's claim that redshift/blueshift records the extent of expansion. It is also the branch form implemented as the Python default `expansion_extent`.

Endpoint Clock Correction

The second redshift-relevant ingredient is Mills's explicit gravitational wavelength statement, Eq. (32.165):

$$\lambda_\infty=\lambda(r)B(r), \qquad B(r)=1+\frac{2Gm_U}{rc^2}. \tag{9}$$

[FORCED] If Eq. (9) is accepted as a wavelength shift law and if two finite r-spheres are compared through the same λ_∞ reference, then the algebraic endpoint clock ratio is

$$\lambda_o B(r_o)=\lambda_e B(r_e), \qquad \mathfrak{z}^{B}_{\mathrm{GUTCP}}(t_e,t_o,z) \iff 1+z=\frac{\lambda_o}{\lambda_e}=\frac{B(r_e)}{B(r_o)}. \tag{10}$$

Removing Eq. (32.165) removes the wavelength-shift law; removing the common λ_∞ comparison removes the finite-radius ratio.

With $r = \mathcal{R}(t)$, the endpoint factor is

$$\boxed{\mathfrak{Z}^{B}_{\mathrm{GUTCP}}(t_e, t_o, z; M_z) \iff 1 + z = \frac{1 + \dfrac{2G\, M_z(t_e)}{c^2 \mathcal{R}(t_e)}}{1 + \dfrac{2G\, M_z(t_o)}{c^2 \mathcal{R}(t_o)}}} \tag{11}$$

where $M_z(t)$ is the mass parameter used in the redshift factor. The most literal Eq. (32.165) choice is $M_z(t) = M_0$. The Eq. (32.158) choice is

$$M_z(t) = m_U(t) = \frac{M_0}{2} \left(1 + \cos\theta(t)\right). \tag{12}$$

[CHOSEN] If the Eq. (32.165) endpoint clock factor is treated as separate from the expansion-extent stretching, the combined redshift relation is

$$\boxed{\mathfrak{Z}^{\mathcal{E}B}_{\mathrm{GUTCP}}(t_e, t_o, z; M_z) \iff 1 + z = \frac{\mathcal{R}(t_o) - \mathcal{R}_{\min}}{\mathcal{R}(t_e) - \mathcal{R}_{\min}} \frac{1 + \dfrac{2G\, M_z(t_e)}{c^2 \mathcal{R}(t_e)}}{1 + \dfrac{2G\, M_z(t_o)}{c^2 \mathcal{R}(t_o)}}.} \tag{13}$$

The Python artifact exposes the denominator choices explicitly: `expansion_extent`, `radius_scale`, `mills_ct_path`, `mills_endpoint`, and combined endpoint modes. The `expansion_extent` mode is the selected GUTCP redshift relation counterpart to the FLRW z parameter; the endpoint factor is a printed Mills correction relation that can be included or audited separately.

Branch Inversion

[FORCED] For fixed t_o, a redshift law is a relation on (t_e, z), not automatically a singleton graph in either direction. For a selected redshift relation $\mathfrak{Z}^{\star}_{\mathrm{GUTCP}}$, define the emission-time fiber

$$\mathcal{T}^{\star}_{z}(z; t_o) = \{t_e < t_o : \ \mathfrak{Z}^{\star}_{\mathrm{GUTCP}}(t_e, t_o, z)\}\,. \tag{14}$$

This set may contain zero, one, or many emission times. A branch condition $\mathfrak{B}_{\mathrm{GUTCP},b}(t_e, t_o, z)$ restricts the fiber to

$$\mathcal{T}^{\star}_{z,b}(z; t_o) = \{t_e < t_o : \ \mathfrak{Z}^{\star}_{\mathrm{GUTCP}}(t_e, t_o, z) \wedge \mathfrak{B}_{\mathrm{GUTCP},b}(t_e, t_o, z)\}\,. \tag{15}$$

Only when this restricted fiber is a singleton is it harmless to write $t_b(z)$ as shorthand for its sole member.

For a radial light-front relation, the corresponding light-travel-distance relation is

$$\mathfrak{D}^{\star}_{\mathrm{GUTCP}}(z, D; t_o) \iff \exists t_e \ [t_e \in \mathcal{T}^{\star}_{z}(z; t_o) \wedge D = c(t_o - t_e)]\,. \tag{16}$$

This relation is the necessary replacement for a single FLRW $D(z)$. Oscillation does not remove distance-redshift solutions; it changes the cardinality of the fibers and therefore makes branch information explicit.

Blueshift Threshold Relations

[FORCED] If t_o is measured from the start of the current expansion and a radial light-front relation is used,

$$\mathfrak{D}_{\mathrm{light}}(t_e, t_o, D) \Longleftrightarrow D = c(t_o - t_e), \tag{17}$$

then the threshold at which the emission event leaves the current-expansion branch is

$$\mathfrak{D}_{\mathrm{enter-blue}}(D_\star, t_o) \Longleftrightarrow D_\star = c\,t_o. \tag{18}$$

For $D < c\,t_o$, the emission time satisfies $0 < t_e < t_o$, so the path lies inside the current expansion branch. For $D > c\,t_o$, the relation gives $t_e < 0$, so a pre-current-expansion segment has entered the path. That is the first threshold at which a contraction/blueshift segment can become part of the redshift relation.

[CHOSEN] This threshold is not the same as the threshold for *net* endpoint blueshift. For the even oscillatory endpoint relations $\mathcal{R}(-t) = \mathcal{R}(t)$ and $\mathcal{E}(-t) = \mathcal{E}(t)$, the endpoint-only neutral condition is $t_e = -t_o$, hence

$$\mathfrak{D}_{\mathrm{net-neutral}}(D_\star, t_o) \Longleftrightarrow D_\star = 2c\,t_o. \tag{19}$$

Under that endpoint-only symmetry, $D > 2c\,t_o$ gives an emission event whose oscillatory radius/extent exceeds the observation value, so the endpoint ratio is net blueshift. Between $c\,t_o$ and $2c\,t_o$, a blueshift segment may enter the path while the endpoint ratio can still be net redshift.

[FORCED] These thresholds are branch-admissibility relations. If an independent observational relation supplies or bounds D, then

$$D < c\,t_o$$

selects the current-expansion branch and excludes branches requiring $t_e < 0$. Conversely,

$$D > c\,t_o$$

excludes a current-expansion-only branch and forces a segmented relation that admits a pre-current-expansion portion of the path.

[CHOSEN] If the symmetric endpoint-only rule is the branch rule being tested, the observed sign of z adds a second pruning condition:

$$\mathfrak{B}_{\mathrm{sign}}(D, t_o, s) \Longleftrightarrow \begin{cases} D < 2c\,t_o, & s = \text{red}, \\ D = 2c\,t_o, & s = \text{neutral}, \\ D > 2c\,t_o, & s = \text{blue}. \end{cases} \tag{20}$$

Thus a positive observed redshift can rule out a candidate branch whose endpoint-symmetric relation already lies beyond the net-blue threshold. This does not make the full distance-redshift relation unique by itself; it removes inadmissible members from the fiber before any singleton branch is selected.

[WEAKEST LINK] The expansion-extent relation has $\mathcal{E}(0) = 0$. A light path that crosses the exact minimum-radius boundary therefore requires an additional bounce or matching relation before the segmented redshift product is physically well defined. The whole-radius relation avoids this zero but includes the finite gravitational-radius offset.

Koksbang–Heinesen FLRW-Consistency Standard

The local reference paper by Koksbang and Heinesen, stored under `docs/cites/` as the model-independent generalized-FLRW consistency PDF, sets a useful external standard for a fair comparison. Its model-independent pipeline does not begin with a cosmological parameter fit. It reconstructs redshift-indexed observable relations, conventionally written in graph shorthand as

$$d_A(z), \qquad d'_A(z), \qquad d''_A(z), \qquad H_\parallel(z), \qquad H'_\parallel(z), \tag{21}$$

where d_A is the angular diameter distance and $H_\parallel$ is the line-of-sight expansion rate

$$H_\parallel = \frac{1}{3}\theta - e^\mu a_\mu + e^\mu e^\nu \sigma_{\mu\nu}. \tag{22}$$

The paper then evaluates generalized FLRW consistency diagnostics over the overlap of the supernova and BAO reconstructions, with the currently useful domain beginning at $z = 0.38$ and extending to order $z \simeq 2$.

[FORCED] BAO is not a direct model-independent d_A datum. The measured object is a clustering-feature relation in angle/redshift space; the usual compressed reductions are ratios or products such as $D_M(z)/r_d$, $H_\parallel(z)r_d$, or $D_V(z)/r_d$. Transverse BAO therefore becomes an angular-ruler constraint only after a sound-horizon/ruler relation and a distance dictionary are admitted. In a GUTCP comparison it belongs on the conditional data-model side, not as the missing GUTCP angular-diameter propagation relation itself.

[FORCED] Mills's page-1579 treatment is a different claimed ruler relation, not a BAO reduction. He takes the present Earth r-sphere radius as

$$r_{\text{sphere}} = 14.02 \times 10^9 \text{ ly}$$

and the corresponding angular view as

$$A_{\text{view}} = 2\pi r_{\text{sphere}} = 88.09 \times 10^9 \text{ ly}.$$

For a CMBR multipole ℓ, the stated sky fraction is $2/\ell$, so the large-structure scale relation is

$$\mathfrak{M}_{\text{CMBR}}(\ell, r_{\text{sphere}}, L) \Longleftrightarrow L = \frac{2}{\ell} A_{\text{view}} = \frac{4\pi r_{\text{sphere}}}{\ell}. \tag{23}$$

This gives $L_{15} = 11.7 \times 10^9$ ly, $L_{135} = 1.3 \times 10^9$ ly, and $L_{700} = 0.25 \times 10^9$ ly, matching the examples Mills assigns to the Hercules–Corona Borealis Great Wall, the Big Ring, and smaller higher-order structures. The relation is therefore a GUTCP/CMBR multipole structure-scale claim. It should not be collapsed into the BAO transverse-scale data product. Testing it against observed arcs or rings still requires the relevant object selection, angular extent, redshift extent, and observational distance dictionary to be stated as relations.

In relation notation, the angular-diameter observation is at least

$$\mathfrak{A}_{\text{obs}}\left(z, d_A, \mathrm{d}A_{\text{emit}}, \mathrm{d}\Omega_{\text{obs}}\right) \Longleftrightarrow d_A^2 = \frac{\mathrm{d}A_{\text{emit}}}{\mathrm{d}\Omega_{\text{obs}}}. \tag{24}$$

This relation can be stated without FLRW. What is still needed for GUTCP is the constitutive relation that connects $(\mathrm{d}A_{\text{emit}}, \mathrm{d}\Omega_{\text{obs}})$ to a selected GUTCP light path, phase, mass convention, and branch.

Two of the diagnostic targets are especially useful for auditing a non-FLRW proposal. The integrated CBL relation is written in singleton-graph shorthand as

$$O(z) = H_{\parallel}(z)^2 D'(z)^2 - 1, \qquad O \xrightarrow{\text{FLRW}} \Omega_{k,0} H_0^2, \tag{25}$$

where D is the distance variable used in the CBL integral relation. The density/ΛCDM diagnostic is

$$M(z) = \frac{2H_{\parallel}(z)^2}{3d_A(z)(1+z)} \left[-d_A'(z) + \left(\frac{2}{1+z} + \frac{H_{\parallel}'(z)}{H_{\parallel}(z)} \right) d_A(z) + d_A''(z) \right], \tag{26}$$

with $M \to \Omega_{m,0} H_0^2$ in ΛCDM under the paper's stated assumptions. The full generalized CBL statistic $C(z)$ similarly combines d_A, $H_{\parallel}$, their derivatives, optical shear/expansion, and Ricci focusing; the FLRW null expectation is $C = 0$.

[FORCED] Applying that standard to GUTCP requires more than a redshift relation. A branch relation must be selected so that

$$t_e \in \mathcal{T}^{\star}_{z,b}(z; t_o),$$

and then the branch-restricted GUTCP quantities must be joined with observable relations. For example,

$$\mathfrak{H}^{\star}_{\parallel,\mathrm{GUTCP},b}(z, H; t_o) \iff \exists t_e \ \left[t_e \in \mathcal{T}^{\star}_{z,b}(z; t_o) \wedge H = H_{\parallel}(t_e) \right], \tag{27}$$

$$\mathfrak{A}^{\star}_{\mathrm{GUTCP},b}(z, d_A; t_o) \iff \exists t_e \ \left[t_e \in \mathcal{T}^{\star}_{z,b}(z; t_o) \wedge \mathfrak{A}_{\mathrm{obs}}(\cdots) \wedge \mathfrak{L}_{\mathrm{GUTCP},b}(\cdots) \right]. \tag{28}$$

Here $\mathfrak{L}_{\mathrm{GUTCP},b}$ denotes the missing GUTCP light-bundle/angular propagation relation. The corrected expansion-extent $\mathfrak{Z}^{\mathcal{E}}_{\mathrm{GUTCP}}$ constrains the redshift-time relation, but it does not by itself constrain $\mathfrak{L}_{\mathrm{GUTCP},b}$.

[FORCED] The frontier relation is open, but it is not unconstrained. Pages 1569, 1577–1579, and the page-1602 footnote give information that any candidate $\mathfrak{L}_{\mathrm{GUTCP},b}$ must either incorporate or explicitly exclude:

$$\Delta T_E(\ell) = C_{\mathrm{effThompson}}\, 77 \,\mathrm{sinc}\Big(\frac{\pi}{140}(\ell + 197) \Big) \ \mu\mathrm{K},$$

$$\Delta T_B(\ell) = r^{1/2} C_{\mathrm{effThompson}}\, 77 \,\mathrm{sinc}\Big(\frac{\pi}{140}(\ell + 197 + 70) \Big) \ \mu\mathrm{K}, \qquad r^{1/2} = \frac{\Delta \aleph}{ct}.$$

Mills identifies these as polarization/lensing constraints: scattering produces E-mode correlation multipoles, gravitational lensing and accelerating spacetime convert E-mode structure into B-mode structure, and the large-scale matter distribution inherits CMBR multipole/lensing patterns. Together with Eq. (23), these pages constrain the possible angular/light-bundle closure but still do not supply the map

$$(t_e, t_o, \text{branch}, \text{source geometry}) \longmapsto (\mathrm{d}A_{\mathrm{emit}}, \mathrm{d}\Omega_{\mathrm{obs}}).$$

Page 1569 adds a separate constraint and a separate risk: Mills uses CMBR structures of about 1° as evidence for nearly flat geometry since the commencement of expansion, but the same sentence ties the geometry claim to a commencement of expansion "about 10 billion years ago." Since the notebook calibration cannot generally accept $t_{\mathrm{now}} = 10^{10}$ yr as an input without overconstraining the H_0/CMB temperature solve, any $\mathfrak{L}_{\mathrm{GUTCP},b}$ derived from that geometry discussion must separate the near-flat/angular-structure claim from the fixed-10-Gyr approximation.

The page-1602 footnote adds an observational-reference constraint. Mills identifies a practical reference for absolute space at rest as a bright celestial body with zero translational velocity within its r-sphere, or with that velocity component corrected. He then proposes a Cepheid of the corresponding calculated age/distance relative to the current r-sphere as an approximate point at rest on a selected r-sphere, and states that changes in angular diameter over the pulsation cycle combined with spectroscopic radial velocity determine distance quasi-geometrically. A candidate $\mathfrak{L}_{\mathrm{GUTCP},b}$ therefore cannot use an arbitrary observer frame: it must either encode this absolute-rest/Cepheid calibration relation or explicitly explain why it is not part of the optical dictionary.

Source	Relation fragment	Constraint on $\mathfrak{L}_{\mathrm{GUTCP}}$	Status
p.1569	BOOMERANG 1° CMBR structures and nearly flat geometry since commencement of expansion	Candidate angular closure must decide whether this is a curvature/flatness constraint, while not importing fixed 10^{10} yr as a calibration input	constraining input with calibration hazard
p.1602 footnote	Absolute-rest reference object and Cepheid angular-diameter plus spectroscopic-radial-velocity distance calibration	Candidate optical dictionary must specify the r-sphere rest-frame/velocity-correction rule and how quasi-geometrical Cepheid distances anchor the scale	constraining input
p.1577, Eq. 32.203	E-mode CMBR multipole pattern from Thompson scattering	Any optical relation including polarization must preserve the stated ℓ-indexed E-mode phase and amplitude convention	constraining input
p.1577, Eqs. 32.204–32.205	B-mode pattern shifted by 70 in ℓ with amplitude ratio $\Delta\aleph/(ct)$	Candidate lensing/accelerating spacetime closure must explain the E-to-B conversion and amplitude ratio	constraining input
p.1578	Gravitational lensing produces rings/arcs and is linked to matter formation patterns	Candidate light-bundle relation must decide how lensing maps photon bundles into observed angular structures	constraining input
p.1579	CMBR multipole scale $L_\ell = 4\pi r_{\mathrm{sphere}}/\ell$	Large-ring/arc comparisons should use the CMBR multipole ruler relation, not BAO transverse scale by default	stated relation

Required relation	Paper standard	Current GUTCP artifact	Fair evaluation
$\mathfrak{Z}_{\mathrm{GUTCP}}$	Observed redshift relation for d_A and $H_\parallel$	$\mathfrak{Z}_{\mathrm{GUTCP}}(t_e, t_o, z)$ stated after choosing denominator/endpoint convention	conditional, branch dependent
$\mathcal{T}_{z,b}$	Restrict redshift fibers to index model quantities	Implemented as multi-valued emission-time fibers	requires branch/window bits
$\mathfrak{H}_{\parallel,\mathrm{GUTCP}}$	Line-of-sight expansion rate relation, not merely a named "Hubble constant"	$H_{\mathcal{E}}$, $H_{\mathcal{R}}$, or Mills H_{ct} joined to a selected branch fiber	partial; convention must be declared
$\partial_z\mathfrak{H}_{\parallel,\mathrm{GUTCP}}$	Redshift derivative relation for C and M	reducible only after a stable branch relation	not yet paper-equivalent
$\mathfrak{A}_{\mathrm{GUTCP}}$	Angular-diameter observational relation reconstructed from supernovae	general relation stated in Eq. (24); GUTCP angular-propagation relation not stated	relation gap
$\partial_z\mathfrak{A}, \partial_z^2\mathfrak{A}$	Required distance-derivative relations	not reducible until $\mathfrak{A}_{\mathrm{GUTCP}}$ is joined to a GUTCP branch/angular relation	relation gap
BAO ruler	BAO constrains D_M/r_d, $H_\parallel r_d$, or D_V/r_d, not bare d_A	can be joined only as a conditional ruler/data relation	not a model-independent GUTCP d_A datum
C, O, M	Diagnostic FLRW/general-spacetime consistency relations with uncertainty bands	cannot be fairly reduced from $H(t)$ and light-travel distance alone	not yet evaluable at full standard

[CHOSEN] The Python artifact therefore exposes a readiness table and branch-specific surrogate observable relations rather than forcing a numerical Koksbang–Heinesen $C/O/M$ reduction. For example, from a selected branch one may state

$$\mathfrak{D}^{\star}_{\mathrm{GUTCP,light},b}(z, D; t_o) \Longleftrightarrow \exists t_e \ \left[t_e \in \mathcal{T}^{\star}_{z,b}(z; t_o) \wedge D = c(t_o - t_e)\right]$$

and a candidate $\mathfrak{H}^{\star}_{\|,\mathrm{GUTCP},b}(z, H; t_o)$. But

$$\mathfrak{D}^{\star}_{\mathrm{GUTCP,light},b} \neq \mathfrak{A}^{\star}_{\mathrm{GUTCP},b}$$

unless an additional GUTCP angular-size propagation law is supplied. Treating light-travel distance as angular diameter distance would import an observational dictionary not present in the audited material.

[FORCED] A fair comparison to the paper is therefore a readiness verdict, not a premature rejection. GUTCP now has candidate redshift relations and branch-restricted fibers. It does not yet meet the paper's full standard because the GUTCP angular-diameter relation, its derivative relations, the optical shear/focusing relations, and an uncertainty relation/pipeline are not yet joined to the current artifact.

[CHOSEN] In MDL terms, the information needed to select a GUTCP branch/window is part of the model description length. That cost should be compared against the information needed by FLRW models to describe the observed departures in angular-size and distance diagnostics. The paper's standard is suitable for that comparison because it asks for observable relations and residuals, not for membership in the FLRW class.

Appendix A: H_0 Target Sweep

The notebook line

```
H0GUTCP = H[tNow]
```

defines a model quantity:

$$H_{0,\mathrm{GUTCP}}(t_{\mathrm{now}}, T_{\mathrm{start}}) = H(t_{\mathrm{now}}; M_0(T_{\mathrm{start}})). \tag{29}$$

It is not, by itself, an independent prediction of the observed Hubble constant. The later Mathematica equality

```
(H0GUTCP/.quantitiesSubs) == QuantityMagnitude[UnitConvert[Subscript[H,0]]]
```

is a calibration condition. To make this explicit, replace the Wolfram entity lookup with a named target:

$$H_{0,\mathrm{GUTCP}}(t_{\mathrm{now}}, T_{\mathrm{start}}) = H_{0,\mathrm{target}}, \tag{30}$$

$$T_U(t_{\mathrm{now}}; T_{\mathrm{start}}) = T_{\mathrm{CMB,obs}}. \tag{31}$$

This is why the original notebook solved for t_{now} instead of using Mills's approximate 10^{10} yr value as an input. If t_{now} is fixed, then T_{start} is the only remaining unknown, but the notebook has two empirical equations to satisfy:

$$H_{0,\mathrm{GUTCP}}(T_{\mathrm{start}}) = H_{0,\mathrm{target}}, \qquad T_U(T_{\mathrm{start}}) = T_{\mathrm{CMB,obs}}.$$

Those two constraints are generally inconsistent for one unknown. Leaving $(t_{\mathrm{now}}, T_{\mathrm{start}})$ free gives a determined two-equation, two-unknown calibration problem rather than an overconstrained one.

The sweep variable is therefore $H_{0,\mathrm{target}}$, commonly entered as `H0TargetKmSecMpc`. Each selected target gives a calibrated pair $(t_{\mathrm{now}}, T_{\mathrm{start}})$. This makes the Hubble-tension scan a model-fitting exercise, not a claim that the GUTCP equations alone selected the empirical H_0.

Appendix B: Observational Relation Gap

The current notebook states time-parametrized relations:

$$H(t), \qquad \text{LightRadius}(t), \qquad T_U(t), \qquad m_U(t), \qquad P_U(t).$$

It does not yet state a complete observational relation

$$t \longleftrightarrow z_{\text{GUTCP}}$$

that is independent of FLRW and reduced by branch constraints. In particular, one should not assume the whole-radius relation

$$\mathfrak{Z}^{\mathcal{R}}_{\text{GUTCP}}(t_e, t_o, z) \iff 1 + z = \frac{\mathcal{R}(t_o)}{\mathcal{R}(t_e)}$$

merely by analogy with FLRW. The denominator audit above derives two GUTCP relations from Mills's own variables: the whole-radius relation Eq. (6), and the expansion-extent relation Eq. (8). The latter is selected here because page 1542 describes redshift/blueshift as time stamps of the *extent of spacetime expansion* between observer and observed.

Eq. (8) states a GUTCP-native redshift relation for a specified emission time, observation time, and redshift value. Oscillation does not imply that the distance-redshift relation has no solution; it implies that the relevant fibers may have multiple members. A complete observational relation must include the relevant branch, observation window, and light path through phases where ancient light may undergo partial blueshifting as well as redshifting. Only after such constraints are joined can external diagnostics be stated as relations:

$$\mathfrak{H}_{\parallel,\text{GUTCP},b}(z, H; t_o) \iff \exists t_e \ \left[t_e \in \mathcal{T}^{\star}_{z,b}(z; t_o) \wedge H = H_{\parallel}(t_e)\right], \tag{32}$$

$$\mathfrak{D}_{\text{GUTCP},b}(z, D; t_o) \iff \exists t_e \ \left[t_e \in \mathcal{T}^{\star}_{z,b}(z; t_o) \wedge D = c(t_o - t_e)\right]. \tag{33}$$

This is the point at which comparisons to non-FLRW observational frameworks, including Heinesen–Clifton style line-of-sight expansion and distance diagnostics, become well-posed. If $\mathcal{T}^{\star}_{z,b}$ is later shown to be a singleton fiber over a chosen domain, then the familiar singleton-graph notation is merely a compression of these relations.

Audit Conclusion

[FORCED] Mills's page 1540–1544 arrow-of-time argument makes spacetime expansion the dynamic variable. Eq. (32.126) then requires the denominator of $H = \dot{R}/R$ to be the compared spacetime scale or extent. The later substitution $R = ct$ in Eq. (32.156) is not forced by that derivation and produces a damped sinc-like $H_{ct}(t)$ from an otherwise oscillatory $\mathcal{R}(t)$.

[CHOSEN] The corrected GUTCP counterpart to FLRW z selected here is the expansion-extent relation

$$\mathfrak{Z}^{\mathcal{E}}_{\text{GUTCP}}(t_e, t_o, z) \iff 1 + z = \frac{\mathcal{R}(t_o) - \mathcal{R}_{\min}}{\mathcal{R}(t_e) - \mathcal{R}_{\min}},$$

because Mills describes redshift/blueshift as a timestamp of the extent of spacetime expansion. The whole-radius relation is also exposed as a convention, but it includes the finite gravitational-radius offset rather than just the expansion extent.

[FORCED] Eq. (11) is separately forced as a finite endpoint clock ratio after accepting Mills's Eq. (32.165), a common λ_∞ reference, and the substitution $r = \mathcal{R}(t)$.

[CHOSEN] The Python artifact defaults to the notebook/numerical period convention $T_U = 2\pi GM_0/c^3$, because that is the convention that reproduces Eq. (32.156) and the quoted $H_0 \simeq 78.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at $t = 10^{10}$ yr when used as Mills's worked example. The calibrated notebook does not use 10^{10} yr as an input because doing so would overconstrain the two empirical conditions. That calibration remains a `mills_ct` calibration unless the denominator convention is changed and the fit is repeated. The Python artifact exposes the printed Eq. (32.149) convention as an option.

[FORCED] Under the Koksbang–Heinesen consistency standard, a fair external evaluation must use angular-diameter, distance-derivative, line-of-sight expansion, and expansion-derivative relations. The current GUTCP artifact states branch-specific redshift fibers, candidate line-of-sight expansion relations, and light-travel-distance relations. It has not yet joined these to the GUTCP angular-diameter relation required by the paper, because the needed light-propagation and angular-size relation has not been stated.

[WEAKEST LINK] The first weak link is Mills's unforced ct denominator in Eq. (32.156). The second is branch selection: Mills does not provide enough material in the audited 2023 pages to select a singleton inverse fiber $\mathcal{T}^{\star}_{z,b}$ for light that crosses the oscillatory history. The third is the angular-diameter relation. The corrected redshift relation can be stated, but a paper-equivalent $C/O/M$ comparison remains conditional until the branch relation, line-of-sight expansion relation, angular-diameter relation, and uncertainty relation/pipeline are joined.